Chem 352L – Grignard Reaction with a Ketone: Synthesis of Triphenylmethanol

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Abstract

Phenylmagnesium bromide was prepared from bromobenzene and magnesium in diethyl ether, allowed to react with benzophenone, and hydrolyzed with aqueous HCl to give triphenylmethanol. The crude product (0.061 g) was examined by thin-layer chromatography (TLC), which showed a major spot at $R_f = 0.703$ (triphenylmethanol) and a minor spot at $R_f = 0.991$ (biphenyl). Two ligroin (petroleum ether) washes removed the nonpolar biphenyl, leaving 0.058 g of purified solid with a single recorded R_f of 0.655. The reagent amounts were 0.720 mmol bromobenzene, 0.864 mmol magnesium, and 0.582 mmol benzophenone, so benzophenone was the limiting reagent and the theoretical yield was 0.1514 g. The percent yield of purified product was 38.3 % (±0.8 percentage points from balance readability alone). The product melted at 161 °C (literature 164 °C), and its infrared spectrum showed an O–H stretch, aromatic C–H and C=C absorptions, and out-of-plane bending bands characteristic of a monosubstituted benzene ring, with no strong carbonyl band to indicate leftover benzophenone. The slightly low melting point and moderate yield point to a small amount of residual impurity and to losses in the Grignard formation, aqueous workup, and ligroin washes.

1 Introduction

Grignard reagents are organomagnesium halides, R−MgX, formed when an organic halide reacts with magnesium metal in an ether solvent. The carbon bonded to magnesium carries a partial negative charge, so a Grignard reagent behaves as a strong carbon nucleophile and as a strong base. This dual reactivity governs both the usefulness of the reaction and the precautions that must be taken. As a nucleophile it adds to the electrophilic carbon of a carbonyl group, forming a new carbon–carbon bond; as a base it is destroyed immediately by any acidic hydrogen, including the hydrogen of water.

In this experiment phenylmagnesium bromide was prepared from bromobenzene and then treated with benzophenone, a diaryl ketone. Addition of the phenyl nucleophile to the carbonyl carbon gives a magnesium alkoxide salt, and aqueous acid protonates this salt to give the tertiary alcohol triphenylmethanol. A competing coupling reaction between the Grignard reagent and bromobenzene (or between phenyl radicals) produces biphenyl, a nonpolar impurity. The crude product was therefore assessed by TLC, purified by washing with ligroin, and characterized by melting point and infrared (IR) spectroscopy. A mass spectrum of triphenylmethanol that was posted in the laboratory was also interpreted.

2 Background and Theory

2.1 Formation of the Grignard Reagent

Magnesium metal is oxidized at its surface by insertion into the carbon–bromine bond:

$$\mathrm{C_6H_5Br + Mg \xrightarrow{Et_2O} C_6H_5MgBr.} \tag{1}$$

Magnesium ribbon is covered by a layer of magnesium oxide that blocks the reaction, so a small crystal of iodine was added to expose fresh metal. The reaction has an induction period; formation of a slightly cloudy solution indicates that the reaction has started, and the purple iodine color fades as the iodine is consumed. Diethyl ether is the standard solvent because it does not react with the Grignard reagent, its oxygen lone pairs coordinate to magnesium and stabilize the reagent, it dissolves the organic starting materials and the organomagnesium intermediate, its low boiling point (34.6 °C) allows a gentle, controllable reflux, and it is easily removed at the end. Because ether evaporates during reflux, solvent was added as needed to keep the liquid level constant.

2.2 Why Water Must Be Excluded

Water is a far stronger acid than the benzene C–H bond, so it protonates the Grignard reagent immediately:

$$\mathrm{C_6H_5MgBr + H_2O \longrightarrow C_6H_6 + Mg(OH)Br.} \tag{2}$$

Any moisture on the glassware therefore destroys nucleophile before it can react with the ketone and lowers the yield, with benzene as the by-product. For this reason the glassware was oven-dried and cooled in a desiccator, and a drying tube of blue (anhydrous) $\mathrm{CaCl_2}$ protected the apparatus from atmospheric moisture.

2.3 Addition to Benzophenone and Acidic Workup

The carbonyl carbon of benzophenone is electrophilic because oxygen is more electronegative than carbon. The phenyl carbon of the Grignard reagent attacks this carbon, the carbon–oxygen π bond shifts onto oxygen, and an insoluble magnesium alkoxide forms:

$$\mathrm{C_6H_5MgBr + (C_6H_5)_2C{=}O \longrightarrow (C_6H_5)_3C{-}OMgBr.} \tag{3}$$

Aqueous acid then protonates the alkoxide and dissolves the magnesium salts in the aqueous layer:

$$\mathrm{(C_6H_5)_3C{-}OMgBr + H_3O^+ \longrightarrow (C_6H_5)_3C{-}OH + Mg^{2+} + Br^- + H_2O.} \tag{4}$$

Overall, one mole of Grignard reagent and one mole of benzophenone give one mole of triphenylmethanol, so the product amount is limited by whichever of these is present in the smaller amount.

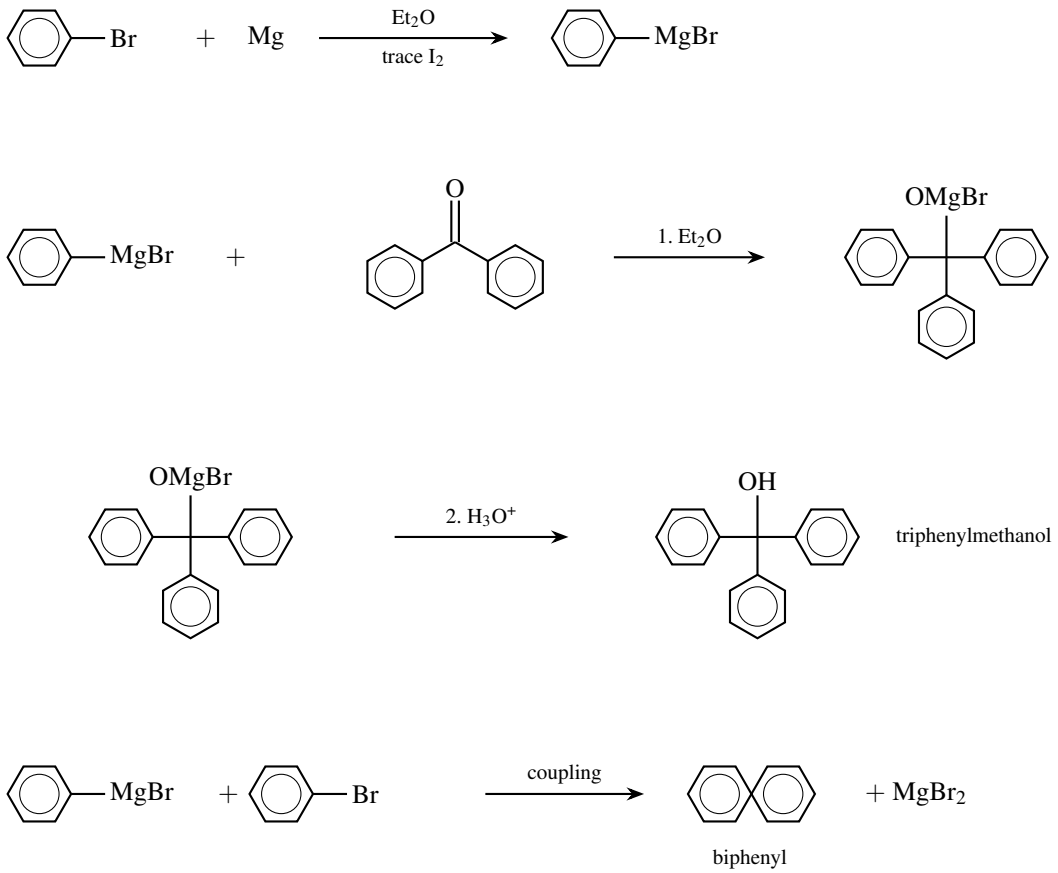


Figure 1: Reaction scheme. Top: formation of phenylmagnesium bromide. Middle: addition to benzophenone to give the magnesium alkoxide, and acidic workup to triphenylmethanol (Ph_3COH). Bottom: the coupling side reaction that forms biphenyl, the impurity removed by the ligroin washes.

2.4 Side Products and Purification

Biphenyl forms when the Grignard reagent couples with unreacted bromobenzene, or when phenyl radicals formed at the metal surface combine. It is favored when bromobenzene is added too quickly or when solvent is allowed to boil away during reflux. Biphenyl is nonpolar, whereas triphenylmethanol has a polar, hydrogen-bonding O–H group. This difference explains both the TLC behavior and the purification. On a polar silica plate the nonpolar biphenyl is held weakly and travels with the methylene chloride (high R_f), while the alcohol is retained more strongly (lower R_f):

$$R_f = \frac{\text{distance traveled by compound}}{\text{distance traveled by solvent front}}. \tag{5}$$

Ligroin is a mixture of nonpolar hydrocarbons, so it dissolves biphenyl readily but dissolves little triphenylmethanol, which can therefore be washed free of biphenyl while remaining as a solid. Benzophenone, which is more polar, is not removed as well by ligroin, so it would be detected by its carbonyl absorption in the IR spectrum.

2.5 Melting Point and Infrared Spectroscopy

A pure crystalline compound melts sharply at its literature temperature, and impurities lower and broaden the melting range. The literature melting point of triphenylmethanol is 164 °C; a range such as 162–165 °C would indicate good purity. In the IR spectrum, triphenylmethanol should show a broad O–H stretch near 3200–3600 $\mathrm{cm^{-1}}$, aromatic $\mathrm{sp^2}$ C–H stretching near 3030 $\mathrm{cm^{-1}}$, aromatic C=C stretching near 1450–1600 $\mathrm{cm^{-1}}$, a C–O stretch of the tertiary alcohol, and strong out-of-plane C–H bending bands near 760 $\mathrm{cm^{-1}}$ and 700 $\mathrm{cm^{-1}}$ for monosubstituted benzene rings. There is no carbonyl group in the product, so a strong band near 1650 $\mathrm{cm^{-1}}$ (benzophenone) would indicate contamination.

2.6 Mass Spectrum of Triphenylmethanol

Triphenylmethanol has the formula $\mathrm{C_{19}H_{16}O}$ and a molar mass of 260.33 $\mathrm{g\,mol^{-1}}$. The spectrum posted in the laboratory shows a base peak at $m/z = 243$, with the protonated molecule $[\mathrm{M+H}]^+$ expected at 261. The peak at 243 is explained by loss of water from the protonated alcohol (equivalently, loss of OH from M), which gives the triphenylmethyl (trityl) cation, $\mathrm{C_{19}H_{15}^+}$. This cation is extremely stable because its positive charge is delocalized over three phenyl rings, which is why ionization of this alcohol gives almost exclusively this fragment (Figure 2).

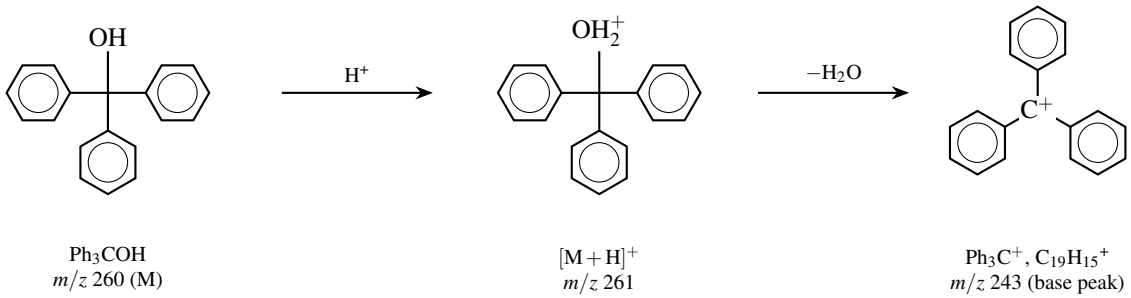


Figure 2: Proposed origin of the base peak at m/z 243: protonation of the alcohol followed by loss of water gives the resonance-stabilized triphenylmethyl cation.

3 Objectives

The objectives of this experiment were to:

1. prepare phenylmagnesium bromide under anhydrous conditions;
2. react the Grignard reagent with benzophenone and hydrolyze the alkoxide to give triphenylmethanol;
3. assess crude purity by TLC and remove biphenyl by washing with ligroin;
4. identify the limiting reagent and calculate the theoretical and percent yields; and
5. characterize the purified product by melting point and IR spectroscopy.

4 Hypothesis

If phenylmagnesium bromide is formed under dry conditions and reacted with benzophenone, then triphenylmethanol will form as the major product, with biphenyl as a minor nonpolar by-product. The product should therefore have a lower TLC R_f than biphenyl, biphenyl should be removed by ligroin washes, and the purified product should melt near 164 °C and show an O–H stretch but no C=O stretch in the IR spectrum.

5 Materials and Equipment

Chemicals: bromobenzene, magnesium ribbon, iodine (one small crystal), benzophenone, anhydrous diethyl ether, 3 M HCl, anhydrous Na_2SO_4, methylene chloride (TLC mobile phase), ligroin (petroleum ether), blue litmus paper, silica TLC plates, and blue (anhydrous) $CaCl_2$ drying agent.

Equipment: 5 mL and 3 mL conical vials, two screw-cap vials, Claisen adapter and head, septum with lid, drying tube, spin vane, 1.0 mL syringe, hot plate, drying oven, desiccator, milligram balance, centrifuge-tube/vial for extraction, Pasteur pipets, beaker, TLC chamber, melting-point apparatus, and an Agilent ATR-IR spectrometer.

6 Safety Considerations

Goggles, a lab coat, and gloves were worn. Diethyl ether is extremely flammable and volatile and can form explosive peroxides, so it was kept in the hood and away from the hot plate and other ignition sources, and the apparatus was never heated as a closed system. Bromobenzene is an irritant and was dispensed in the hood. Iodine is harmful if inhaled, swallowed, or in contact with skin and eyes, can damage organs (including the thyroid) after repeated exposure, and is toxic to aquatic life, so it must not go down the drain. Magnesium is flammable as a fine powder. Benzophenone, methylene chloride, and ligroin are irritants, and the latter two were used only in the hood. Hydrochloric acid (3 M) is corrosive and was added dropwise. Hot oven glassware was handled with tongs and cooled in the desiccator, and spills were cleaned up promptly.

7 Methods and Procedure

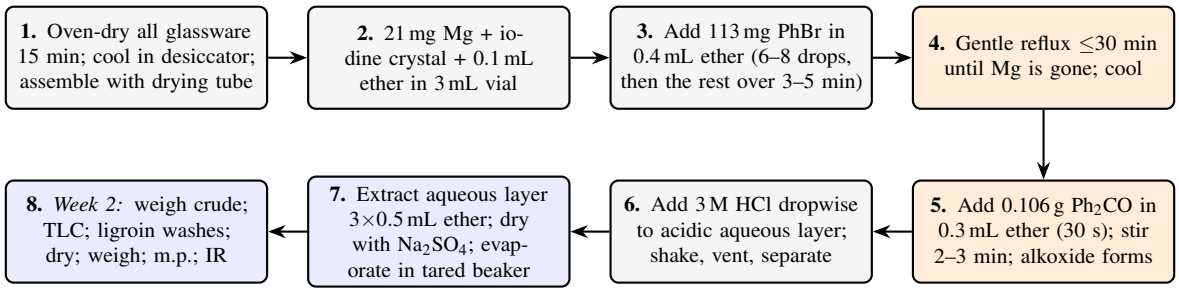


Figure 3: Workflow of the two-week synthesis. Steps 1–7 were completed on October 6; step 8 was completed on October 13.

7.1 Week 1: Grignard Reaction, Workup, and Extraction (October 6)

The 5 mL and 3 mL conical vials, two screw-cap vials, Claisen head, and drying tube were dried in the oven for 15 min and cooled in a desiccator. A dried, tared 3 mL vial containing a spin vane was attached to the Claisen adapter with a very small amount of grease; one Claisen head was connected to a drying tube of blue $CaCl_2$ and the other was fitted with a septum and lid. About 5 mL of diethyl ether was placed in the dried 5 mL vial as a reserve. Magnesium ribbon (21 mg) cut into very small pieces, a small crystal of iodine, and about 0.1 mL of ether were added to the reaction vial. In a dry screw-cap vial, 113 mg (76 µL) of bromobenzene was dissolved in about 0.4 mL of ether and drawn into a 1.0 mL syringe, which was inserted through the septum without depressing the plunger.

With slow stirring, 6–8 drops of the bromobenzene solution were added to initiate the reaction (slight cloudiness), and the rest was then added over 3–5 minutes. The empty vial was rinsed with about 0.3 mL of ether, and the rinse was added in one portion. The mixture was heated at a gentle reflux (hot plate setting about 3) until the magnesium had disappeared, for no more than 30 min, with ether added as needed to keep the solvent level constant, and then cooled to room temperature. Benzophenone (0.106 g) was dissolved in about 0.3 mL of anhydrous ether in a dry capped vial, drawn into the syringe, and added through the septum over about 30 s so that the mixture refluxed no more than gently. The vial was rinsed with 0.3 mL of ether and the rinse was added. After 2–3 min of stirring and cooling, the Claisen head was removed and the vial was capped.

The magnesium alkoxide was hydrolyzed by dropwise addition of 3 M HCl with stirring until the lower aqueous layer tested acidic on blue litmus paper. The spin vane was removed, the vial was capped, shaken, and vented, and the layers were allowed to separate (ether on top). The bottom aqueous layer was transferred to a clean 5.0 mL vial and extracted with three 0.5 mL portions of ether, each time combining the top ether layer with the earlier organic layers. The combined ether solution was dried with anhydrous Na_2SO_4 until clear, transferred to a tared beaker (49.547 g), and left in the drawer to evaporate for a week.

7.2 Week 2: Purification and Characterization (October 13)

The beaker containing the crude product was weighed. The crude product was analyzed by TLC on silica with methylene chloride as the mobile phase. Two milliliters of ligroin was added to the crude solid, the liquid was pipetted off, and the washing was repeated with a second portion. The remaining solid was checked by TLC, air-dried, and weighed. Its melting point was measured and an ATR-IR spectrum (16 scans, resolution 4 cm^{-1}, range 650–4000 cm^{-1}) was recorded and labeled with the diagnostic peaks.

8 Results

8.1 Recorded Masses and Reagent Table

Table 1: Masses recorded in the laboratory notebook.

Quantity	Value	Notes
Magnesium ribbon	0.021 g	Procedure suggests 18 mg; 25 mg was crossed out
Bromobenzene	113 mg (76 µL)	Density 1.495 g mL^{-1}
Benzophenone	0.106 g	Procedure suggests 105 mg
Tared reaction vial	20.191 g	Vial + spin vane: 20.816 g
Tared beaker	49.547 g	Before evaporation of ether
Beaker + crude product	49.608 g	After ether evaporated (one week)
Crude product	0.061 g	49.608 g − 49.547 g
Purified triphenylmethanol	0.058 g	After two ligroin washes (0.053 g was crossed out)

Table 2: Reagent and product table. Literature values for benzophenone and biphenyl are given for reference only and were not measured.

Compound	MW (g/mol)	Amount	mmol	Lit. m.p. (°C)	Exp. m.p. (°C)
Bromobenzene	157.01	113 mg	0.720	–	–
Magnesium	24.305	21 mg	0.864	–	–
Iodine	253.81	small crystal	–	–	–
Benzophenone	182.22	106 mg	0.582	48–49	–
Triphenylmethanol	260.33	58 mg	0.223	164	161
Biphenyl (by-product)	154.21	–	–	69–71	–

8.2 TLC Results

On the crude product, the solvent front traveled 5.55 cm; the large spot was at 3.90 cm and the small spot at 5.50 cm. On the purified product, the solvent front traveled 6.11 cm and the spot was at 4.0 cm.

Table 3: TLC results on silica with methylene chloride as the mobile phase.

Plate	Spot	Distance (cm)	R_f	Lit. R_f (manual)
Crude (front 5.55 cm)	Triphenylmethanol (large)	3.90	0.703	0.6
	Biphenyl (small)	5.50	0.991	0.9
Purified (front 6.11 cm)	Triphenylmethanol	4.0	0.655	0.6

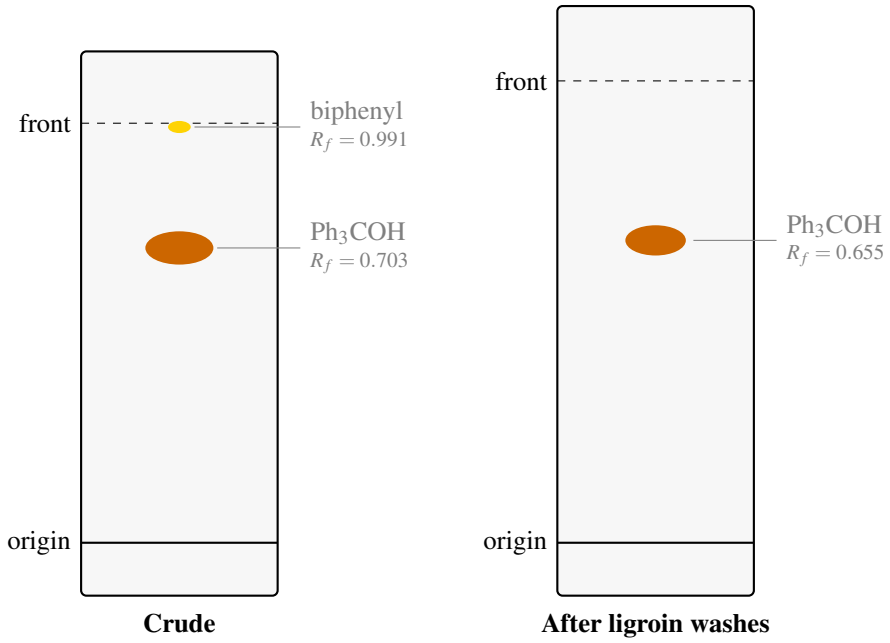


Figure 4: TLC plates drawn to scale (1 cm on paper = 1 cm on the plate) from the recorded distances. After two ligroin washes only the triphenylmethanol spot is shown, because only one R_f was recorded for the purified solid.

8.3 Yield

Table 4: Mass and yield of product at each stage, relative to the theoretical yield based on the limiting reagent (benzophenone, 0.582 mmol).

Stage	Mass (g)	mmol Ph_3COH	Percent of theoretical
Theoretical (benzophenone limiting)	0.1514	0.582	100
Crude (before ligroin)	0.061	0.234	40.3
Purified (after ligroin)	0.058	0.223	**38.3**

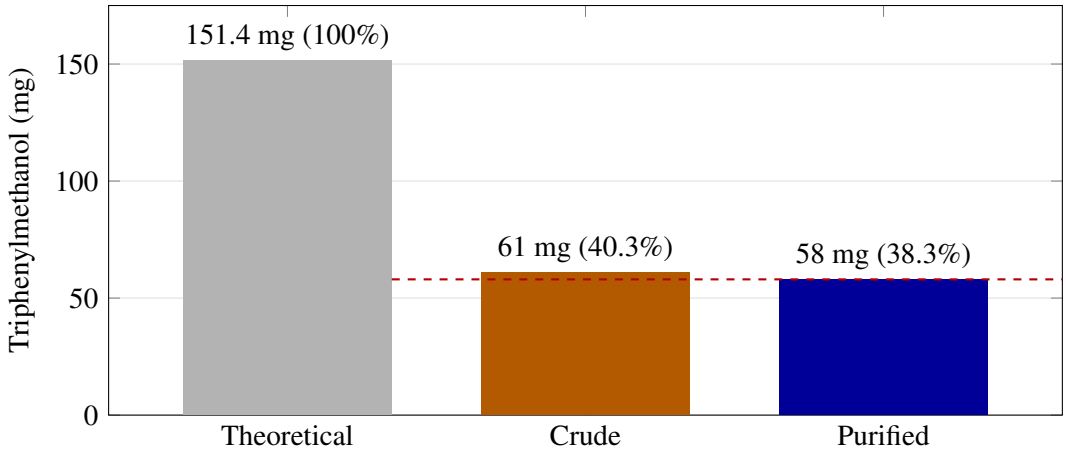


Figure 5: Theoretical, crude, and purified masses of triphenylmethanol. The 3 mg difference between crude and purified (dashed line) is the material removed in the ligroin washes, mostly biphenyl.

8.4 Melting Point

The purified product melted at 161 °C, which is 3 °C below the literature value of 164 °C (1.8 % lower). Only a single temperature was recorded, not a range.

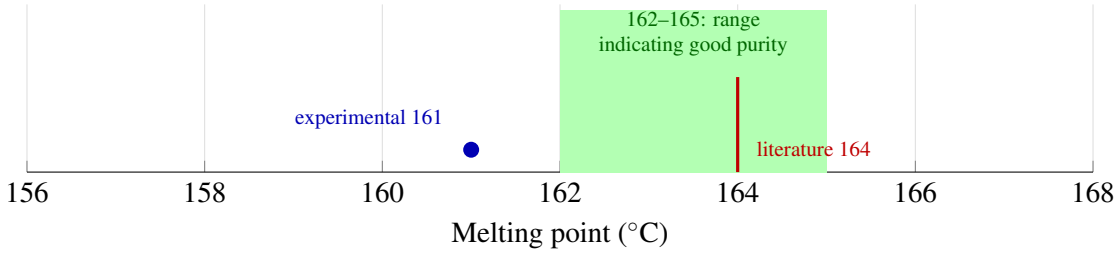


Figure 6: Experimental melting point compared with the literature value and the 162–165 °C window described in the manual.

8.5 Infrared Spectrum

The ATR spectrum (Agilent, 10/13/2026, 16 scans, resolution 4 cm^{-1}) had the diagnostic bands labeled in the notebook: a broad O–H stretch above 3400 cm^{-1}, a weak sp^2 C–H stretch near 3050 cm^{-1}, a weak C=C band near 1600 cm^{-1}, and two strong bands at the low-wavenumber end that were circled as the monosubstituted-benzene substitution pattern (near 760 cm^{-1} and 700 cm^{-1}). The printout shows no strong band in the carbonyl region near 1650 cm^{-1}. Band positions were read from the printed plot and are accurate only to roughly ± 30 cm^{-1}.

Table 5: Diagnostic IR absorptions of triphenylmethanol (positions read from the printed spectrum are approximate).

Assignment	**Approx. position (cm^{-1})**	**Comment**
O–H stretch (alcohol)	3400–3450	Broad, hydrogen-bonded; expected 3200–3600
sp^2 C–H stretch (aromatic)	≈3050	Expected near 3030
C=C stretch (aromatic ring)	≈1600	Weak band, circled in the notebook
C–O stretch (tertiary alcohol)	≈1000–1200	Expected for the C–OH bond
Out-of-plane C–H bend	≈760 and ≈700	Monosubstituted benzene (“substitution”)
C=O stretch (benzophenone)	none observed	Would be strong near 1650

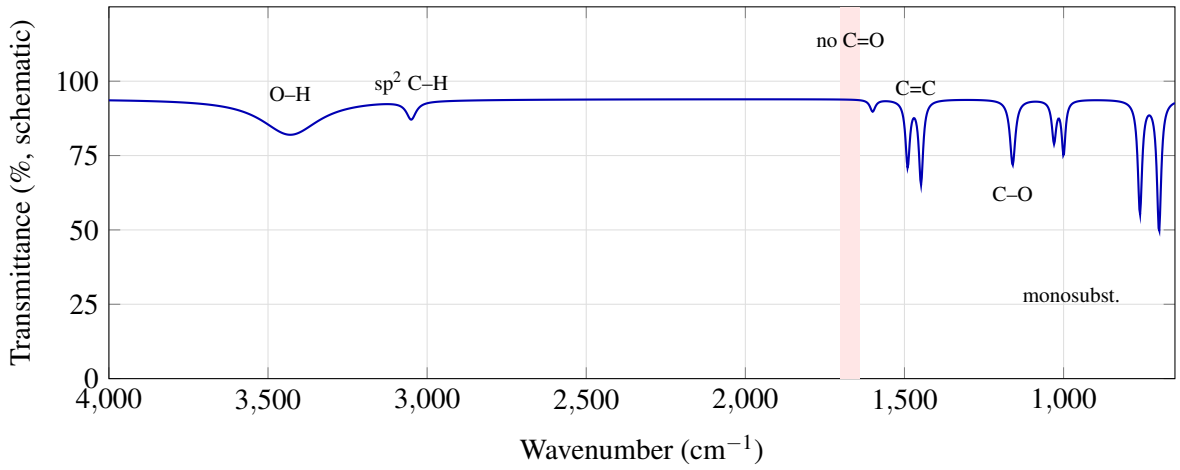


Figure 7: Schematic IR spectrum of triphenylmethanol based on the labeled notebook printout; peak positions are approximate and intensities are illustrative. The shaded band marks the carbonyl region where benzophenone would absorb strongly (not observed). This is not a digitized instrument output.

9 Calculations

9.1 Moles of Each Reagent

$$n_{\mathrm{PhBr}} = \frac{0.113\,\mathrm{g}}{157.01\,\mathrm{g\,mol^{-1}}} = \boxed{0.720\,\mathrm{mmol}} \qquad (76\,\mu\mathrm{L} \times 1.495\,\mathrm{g\,mL^{-1}} = 113.6\,\mathrm{mg}\ \text{confirms the mass}),$$

$$n_{\mathrm{Mg}} = \frac{0.021\,\mathrm{g}}{24.305\,\mathrm{g\,mol^{-1}}} = \boxed{0.864\,\mathrm{mmol}},$$

$$n_{\mathrm{Ph_2CO}} = \frac{0.106\,\mathrm{g}}{182.22\,\mathrm{g\,mol^{-1}}} = \boxed{0.582\,\mathrm{mmol}}.$$

The notebook entry “0.72 mol” for bromobenzene is 0.72 mmol.

9.2 Limiting Reagent

Bromobenzene and magnesium react 1:1 to form the Grignard reagent; bromobenzene is the smaller amount, so at most 0.720 mmol of PhMgBr can form (magnesium is in $0.864/0.720 = 1.20$-fold excess). The Grignard reagent and benzophenone then react 1:1:

$$n_{\text{PhMgBr, max}} = 0.720\,\text{mmol} > n_{\text{Ph}_2\text{CO}} = 0.582\,\text{mmol}.$$

Benzophenone is therefore the **limiting reagent**, and the Grignard reagent is in $0.720/0.582 = 1.24$-fold excess (provided it forms completely).

9.3 Theoretical Yield

$$m_{\text{theor}} = 0.582\,\text{mmol}\left(\frac{260.33\,\text{g mol}^{-1}}{1000}\right) = \boxed{0.1514\,\text{g}}.$$

9.4 Percent Yield

$$\begin{aligned} n_{\text{Ph}_3\text{COH}} &= \frac{0.058\,\text{g}}{260.33\,\text{g mol}^{-1}} = 0.223\,\text{mmol} \quad (\text{notebook: } 2.2\times10^{-4}\ \text{mol}),\\ \%\ \text{yield} &= \frac{0.058\,\text{g}}{0.1514\,\text{g}}\times 100 = \boxed{38.3\,\%},\\ \%\ \text{yield (crude)} &= \frac{0.061\,\text{g}}{0.1514\,\text{g}}\times 100 = 40.3\,\%. \end{aligned}$$

The percent yield is based on benzophenone, because it limits the amount of product that can form.

9.5 Uncertainty in the Yield

Taking a balance readability of ± 0.001 g for the product and benzophenone masses,

$$\frac{u_{\text{yield}}}{\text{yield}} = \sqrt{\left(\frac{0.001}{0.058}\right)^2 + \left(\frac{0.001}{0.106}\right)^2} = 0.020, \qquad u_{\text{yield}} = 0.020\times 38.3 = 0.8\,\%.$$

The yield is therefore 38.3 ± 0.8 % from weighing uncertainty alone. If the manual value of 105 mg benzophenone were used instead of 106 mg, the yield would be 38.7 %.

9.6 Yield Based on Other Reagents

For comparison, the percent yield based on the other reagents would be

$$\frac{0.223}{0.720}\times 100 = 31.0\,\%\ (\text{based on bromobenzene}), \qquad \frac{0.223}{0.864}\times 100 = 25.8\,\%\ (\text{based on Mg}).$$

These are lower bounds on how much of each reagent was converted to product, but they are not the correct yield because benzophenone limits the product.

9.7 Mass Lost in the Ligroin Washes

$$m_{\text{lost}} = 0.061\,\text{g} - 0.058\,\text{g} = 0.003\,\text{g}, \qquad \frac{0.003}{0.061} \times 100 = 4.9\,\% \text{ of the crude mass.}$$

If all 3 mg were biphenyl ($154.21\,\text{g mol}^{-1}$), this would correspond to 0.019 mmol, only about 2.7 % of the bromobenzene.

9.8 R_f Values and Melting-Point Comparison

$$R_f(\text{Ph}_3\text{COH, crude}) = \frac{3.90}{5.55} = 0.703, \qquad R_f(\text{biphenyl}) = \frac{5.50}{5.55} = 0.991,$$
$$R_f(\text{purified}) = \frac{4.0}{6.11} = 0.655,$$
$$\%\text{ difference (m.p.)} = \frac{|161 - 164|}{164} \times 100 = 1.8\,\%.$$

(The notebook gives 0.656 for the purified value; $4.0/6.11 = 0.6547$.)

10 Data Analysis

10.1 Stoichiometry of the Reagents

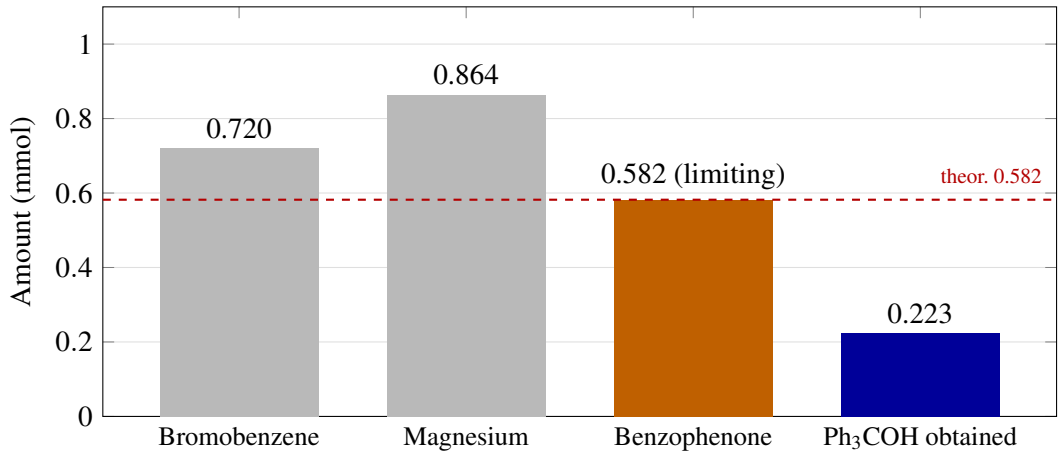


Figure 8: Moles of each reagent and of purified product. Benzophenone (orange) limits the product; the product obtained is 38.3 % of the theoretical amount.

Table 6: Mole ratios of the reagents relative to benzophenone (1.00).

Reagent	**mmol**	**Equivalents vs. Ph_2CO**
Benzophenone (limiting)	0.582	1.00
Bromobenzene	0.720	1.24
Magnesium	0.864	1.49

Using 21 mg of magnesium rather than the suggested 18 mg (0.741 mmol) increased the excess of magnesium over bromobenzene from 1.03 to 1.20 equivalents. The pre-lab calculation for 25 mg (1.03 mmol) shows the same principle: because bromobenzene is the limiting source of phenyl groups in the Grignard step, extra magnesium (here about 0.14 mmol, 3.5 mg) cannot give more product and may remain as unreacted metal.

10.2 Sources of Product Loss

The actual amount of product (0.223 mmol) shows that at least 31 % of the bromobenzene was converted to productive Grignard reagent and added to the ketone. The yield was 38.3 %, so several processes could be responsible for the remaining 61.7 % of the benzophenone:

- **Destruction of the Grignard reagent** by traces of water on the glassware, in the solvent, or from the air, which gives benzene and removes nucleophile;
- **Biphenyl formation** (coupling), which consumes phenyl groups and bromobenzene;
- **Incomplete formation** of PhMgBr, for example if the reaction was slow to initiate or the reflux was shortened or stopped, or if ether was lost by evaporation;
- **Mechanical losses** during transfers between vials, the aqueous extraction (a product that stays in the aqueous layer or on glass), and evaporation of ether in the beaker; and
- **Loss in the ligroin washes**: 0.003 g (4.9 % of the crude) left in solution, as dissolved biphenyl and a little dissolved product.

Because the scale of the reaction is very small (58 mg of product), even a few milligrams lost in a transfer corresponds to several percent of the yield. The notebook does not record observations of reaction initiation, the pink color, or the alkoxide solid, so it is not possible to say from the record which of these effects was the largest.

11 Discussion

Reaction and product identity. The synthesis gave a white solid whose TLC behavior, melting point, and IR spectrum are consistent with triphenylmethanol. The crude mixture showed a large spot at $R_f = 0.703$ and a small spot at $R_f = 0.991$. The lower spot is triphenylmethanol and the upper spot is biphenyl, as predicted by polarity: the hydroxyl group of triphenylmethanol interacts with the polar silica gel, whereas biphenyl has no polar group and travels almost with the solvent front. The measured R_f values (0.703 and 0.991) are somewhat higher than the manual's values (0.6 and 0.9), but both are in the correct order and spacing, and R_f

depends on the plate, chamber saturation, and spotting, so exact agreement is not expected. After two ligroin washes only one R_f (0.655) was recorded for the product, consistent with removal of biphenyl.

Yield. The percent yield, 38.3 %, was calculated from benzophenone, the limiting reagent. This is a moderate yield that is typical of a small-scale Grignard reaction in which moisture and mechanical losses are significant. The 3 mg lost in the ligroin washes is a small fraction (4.9 %) of the crude product, which indicates that the crude material was already mostly triphenylmethanol and that the loss in the purification step was not the main reason for the moderate yield. The main losses therefore probably occurred earlier in the procedure (Section 10.2), but the recorded data cannot distinguish among them.

Melting point and purity. The melting point of 161 °C is 3 °C below the literature value and below the 162–165 °C range that indicates good purity. A depressed melting point is the usual sign of impurity; likely contaminants are a small amount of biphenyl or benzophenone, residual ligroin or ether, or incomplete drying. Because only a single temperature (not a range) was recorded, the breadth of melting cannot be judged, and thermometer calibration or heating rate may also contribute. The IR spectrum supports a largely pure product: the O–H band confirms the alcohol, the aromatic C–H and C=C bands and the strong out-of-plane bends near 760 cm^{-1} and 700 cm^{-1} confirm monosubstituted benzene rings, and the absence of a strong carbonyl band near 1650 cm^{-1} shows that no appreciable benzophenone remains. IR is, however, not very sensitive to small amounts of biphenyl, which has no distinct functional-group bands, so the melting point and TLC are more informative about that impurity.

Mass spectrum. The posted mass spectrum agrees with the structure: the base peak at $m/z = 243$ corresponds to loss of OH from the alcohol (equivalently, loss of water from $[\mathrm{M}+\mathrm{H}]^+$ at 261), giving the very stable triphenylmethyl cation. The molecular ion at 260 is therefore expected to be weak or absent.

Hypothesis. The hypothesis was largely supported: triphenylmethanol was the major product, biphenyl was a minor nonpolar by-product that appeared at higher R_f and was removed by the ligroin washes, and the product showed an O–H band and no C=O band. The melting point, however, was 3 °C lower than predicted.

Improvements. To raise the yield, the glassware and ether should be kept scrupulously dry, the reagents should be weighed so that benzophenone is not limiting unless that is intended, the bromobenzene should be added slowly to minimize biphenyl, and the solvent level should be maintained during reflux. Observations during the reaction (initiation, color change, solid formation) should be recorded, the melting range should be recorded rather than a single value, and a second TLC of the purified product beside a biphenyl standard would confirm one-spot purity.

12 Conclusion

Phenylmagnesium bromide was prepared from 0.720 mmol of bromobenzene and 0.864 mmol of magnesium and was allowed to react with 0.582 mmol of benzophenone, which was the limiting reagent. After acidic workup, extraction, and two ligroin washes, 0.058 g of triphenylmethanol was isolated, a percent yield of $\boxed{38.3\%}$ (± 0.8 percentage points from weighing uncertainty) based on the theoretical yield of 0.1514 g. TLC showed the expected polarity order, with triphenylmethanol ($R_f = 0.703$ in the crude) retained more strongly than biphenyl ($R_f = 0.991$), and the ligroin washes removed the biphenyl. The melting point of 161 °C was slightly below the literature value of 164 °C, indicating a small amount of impurity, and the IR spectrum showed O–H, aromatic, and monosubstituted-benzene absorptions with no carbonyl band. The moderate yield most likely reflects loss of Grignard reagent to moisture or coupling and mechanical losses at the small scale.

A Pre-Lab Questions

Week 1

1. Why must the glassware be dry? What is the product of the reaction with water? Water reacts rapidly with phenylmagnesium bromide, protonating the nucleophilic phenyl carbon before it can react with benzophenone and thereby lowering the yield. The products are benzene and magnesium hydroxybromide: $C_6H_5MgBr + H_2O \longrightarrow C_6H_6 + Mg(OH)Br$.

2. Why is ether a good solvent? (i) It is relatively unreactive toward the Grignard reagent. (ii) Its oxygen coordinates to magnesium and stabilizes the reagent. (iii) It dissolves bromobenzene, benzophenone, and the organomagnesium intermediate. (iv) Its low boiling point allows gentle reflux. (v) It is easily evaporated after the reaction.

3. What if 25 mg **of Mg is used instead of** 18 mg**?** Twenty-five milligrams is 1.03 mmol, whereas the 0.720 mmol of bromobenzene needs only 0.720 mmol of Mg, so about 0.31 mmol (7.5 mg) of magnesium could remain unreacted. Bromobenzene is the limiting reagent in the Grignard step, so extra magnesium does not form more product; some may be consumed by side reactions or moisture, but not all of it will dissolve during reflux.

4. Iodine pictograms. The health-hazard symbol means iodine may cause serious or long-term effects, including damage to organs such as the thyroid after repeated exposure and respiratory effects. The exclamation mark means it is harmful or irritating if inhaled, swallowed, or in contact with skin and eyes. The environment symbol means it is toxic to aquatic organisms and must not be released into drains.

5. Why is triphenylmethanol not very water soluble? Its single hydroxyl group can hydrogen bond with water, but the molecule also has three large, nonpolar, hydrophobic phenyl rings. The nonpolar part is much larger than the polar part, so the molecule interacts poorly with water.

Week 2

1. Why a lower R_f than biphenyl? Triphenylmethanol has a polar O–H group that interacts strongly with the polar silica gel, whereas biphenyl has no polar functional group and has a greater affinity for the nonpolar mobile phase.

2. What does 162–165 °C **indicate?** A narrow range that brackets the 164 °C literature value would indicate a pure product.

3. Are "ether" and "petroleum ether" the same? No. Ether means diethyl ether, $(C_2H_5)_2O$, whereas petroleum ether (ligroin) is a mixture of aliphatic hydrocarbons of varying chain length; it is not an ether.

4. NMR of triphenylmethanol. The posted ^{1}H NMR spectrum shows a large signal at about 7.25 ppm for the 15 aromatic hydrogens and a very small signal for the single O–H hydrogen (its position depends on solvent and concentration). By symmetry, the three phenyl rings are equivalent and the predicted ^{13}C spectrum has five signals: the carbinol carbon near 82 ppm, the ipso carbon near 147 ppm, and ortho, meta, and para carbons between about 127–128 ppm (Figure 9).

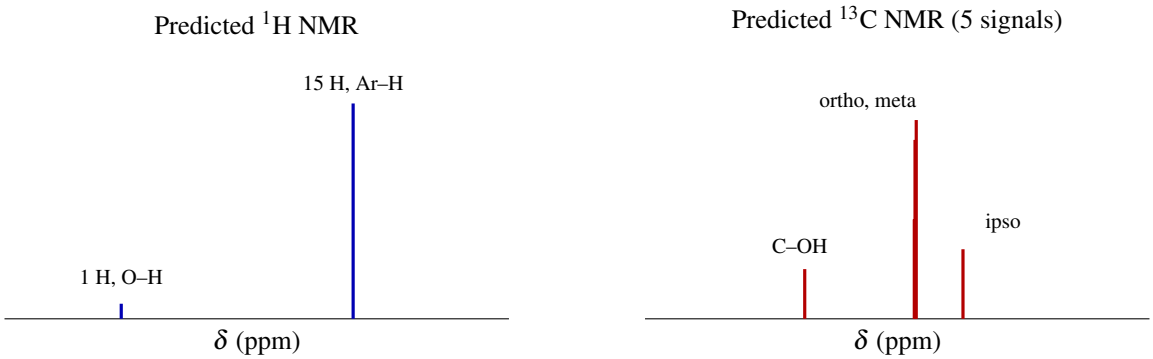


Figure 9: Sketch of the predicted NMR spectra of triphenylmethanol (not measured). Chemical shifts are approximate and the peak heights are only qualitative.

5. IR peaks. Aromatic C–H stretch near 3030 cm^{-1}; O–H stretch of the alcohol at 3200–3600 cm^{-1}; aromatic C=C near 1450 to 1600 cm^{-1}; and the C–O stretch of the alcohol near 1000–1200 cm^{-1}.

Waste Disposal

Diethyl ether, ligroin, and other organic liquids were placed in the organic waste container, and the methylene chloride from TLC in the halogenated organic waste. The acidic aqueous layer, which contains magnesium salts, was discarded as directed by the instructor, and iodine-containing material was not poured down the drain. Used drying agent, TLC plates, and contaminated tissues were placed in the solid chemical waste.

References

1. BYU–Idaho Chemistry Laboratory. *Grignard Reaction with a Ketone: Triphenylmethanol*, Chem 220L (v. 2018).

2. Pavia, D. L.; Lampman, G. M.; Kriz, G. S.; Engel, R. G. *Introduction to Organic Laboratory Techniques: A Small-Scale Approach*, 3rd ed.; Brooks/Cole: Belmont, CA, 2011.

3. Clayden, J.; Greeves, N.; Warren, S. *Organic Chemistry*, 2nd ed.; Oxford University Press: Oxford, 2012.

4. Silverstein, R. M.; Webster, F. X.; Kiemle, D. J.; Bryce, D. L. *Spectrometric Identification of Organic Compounds*, 8th ed.; Wiley: Hoboken, NJ, 2015.